

CUSTOMER SEGMENTATION USING K-MEANS CLUSTERING

1. Introduction

Customer segmentation is the process of dividing customers into groups based on behavior and demographic information. This project uses the Mall Customer Dataset to group customers using the K-Means Clustering algorithm. The analysis helps businesses understand customer spending behavior and improve marketing strategies.

2. Objective

- Perform data preprocessing
- Encode categorical variables
- Apply K-Means clustering
- Visualize customer segments
- Generate business insights

3. Dataset Description

The dataset contains customer details such as Gender, Age, Education, Marital Status, Annual Income, and Spending Score. These features are used to identify customer behavior patterns.

4. Tools and Libraries Used

Python, Pandas, Matplotlib, Scikit-learn, KMeans, StandardScaler, and LabelEncoder.

5. Methodology

The dataset was loaded using Pandas. Missing values were removed and categorical columns such as Gender, Education, and Marital Status were encoded using LabelEncoder. Feature scaling was performed using StandardScaler before applying K-Means clustering.

6. Elbow Method

The Elbow Method was used to determine the optimal number of clusters. The WCSS values were calculated for different cluster counts, and 5 clusters were selected.

7. K-Means Clustering

K-Means clustering was applied with 5 clusters. Customers with similar characteristics were grouped together based on spending behavior and income.

8. Visualization

A scatter plot was created using Annual Income and Spending Score to visualize customer segments. Different colors represent different customer groups.

9. Output Generated

- output.csv – Final clustered dataset
- elbow.png – Elbow method graph
- clusters.png – Customer segmentation graph

10. Business Insights

High-income customers with high spending are premium customers. Low-spending customers may require promotional offers. Customer segmentation helps improve marketing and business decisions.

11. Conclusion

This project successfully performed customer segmentation using K-Means clustering. The analysis grouped customers based on their spending patterns and demographic details, helping businesses understand customer behavior more effectively.

Submitted By:

Amala Binisha C

M.Sc. Data Science